Name_____ Date_____ Period_____

LESSON 8-4 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Writing Inequalities

Maryland Standard 6.EE.B.8 · Learning target – I can write an inequality that describes a constraint in a real situation.

REMEMBER

- An **inequality** compares amounts using <, >, ≤, or ≥.
- **more than**: > **less than**, **fewer than**: <
- **at least**: ≥ **at most**, **no more than**: ≤
- "At least" and "at most" **include** the boundary.

WORKED EXAMPLE

A food pantry needs at least 40 volunteers on Saturday. Write an inequality for v, the number of volunteers.

Step 1 "At least 40" means 40 or more, so 40 is included.

Step 2 Use ≥ : $v \geq 40$. Test: 40 and 52 work; 38 does not.

Answer $v \geq 40$

WORD BANK

at least – that number or more (≥)

at most / no more than – that number or less (≤)

more than – greater, not equal (>)

fewer than / below – smaller, not equal (<)

PRACTICE

show your work · label units

1. Write an inequality for each phrase. Use x.

Step 1 Find the key words.

Step 2 Is the number itself allowed? Yes: ≤ or ≥. No: < or >.

a. more than 12 _____

b. at most 30 _____

c. fewer than 8 _____

d. at least 15 _____

e. no more than 6 _____

f. greater than −3 _____

g. below 0 _____

h. a maximum of 100 _____

2. Write an inequality for each sign. Use the variable given.

Sign	Inequality
Riders must be at least 54 inches tall. (*h*)	
Speed limit: 35 miles per hour. (*s*)	
Buy more than 10 items to get a discount. (*n*)	
Fewer than 120 people may be in this room. (*p*)	

3. **City Bus.** A city bus can carry **no more than** 48 riders. Write an inequality for *r*, the number of riders. Can the bus carry 48 riders? Can it carry 50 riders? Explain.

Step 1 "No more than" means that number or less.

Step 2 Test 48. Then test 50.

Inequality_____

4. **Reverse It.** Write a real-world situation for $t < 60$. Name two solutions.

Step 1 $t < 60$ means t is less than 60.

Step 2 Think of something that must stay under 60, like minutes.

Situation and solutions_____

5. **Same or Different?** Jordan says "x is at most 10" and "x is less than 10" mean the same thing. Is he right? Use the number 10 to explain.

Frame *Jordan is ___ because 10 ___ a solution of $x \leq 10$, but 10 ___ a solution of $x < 10$.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Look for a sign with a limit (speed limit, height rule, room capacity). Write an inequality for it and tell someone at home what it means.

Name_____ Date_____ Period_____

LESSON 8-4 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Representing Inequalities

Maryland Standard 6.EE.B.8 · Learning target – I can represent an inequality with a set of solutions and describe them.

REMEMBER

- **Open dot**: boundary **not** included (< or >).
- **Closed dot**: boundary **is** included (≤ or ≥).
- Arrow: **left** for < and ≤, **right** for > and ≥.

WORKED EXAMPLE

Graph $x \leq 3$.

Step 1 ≤ includes 3, so draw a closed dot at 3.

Step 2 Shade left: the solutions are 3 and less.

Answer **Closed dot at 3, arrow left**

WORD BANK

open circle – the number is not a solution

closed circle – the number is a solution

arrow – points to all the solutions

PRACTICE

use the number lines · show your work

1. Write the inequality shown on the number line. Use x.

Step 1 Find the number at the circle. Is it open or closed?

Step 2 Which way does the arrow point? Right means greater.

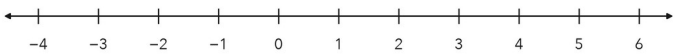


Inequality_____

2. Graph $x \geq 2$ on the number line. Then name two solutions.

Step 1 ≥ includes 2, so the circle is closed.

Step 2 Draw the arrow toward the greater numbers.



Two solutions_____

3. Each graph is described in words. Write the inequality for each graph. Use x.

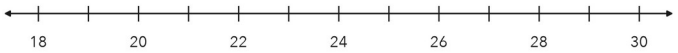
a. open dot at 3, arrow right _____

b. closed dot at −2, arrow left _____

4. Water Bottle. A bottle holds at most 24 ounces of water. Write an inequality for w, the ounces of water. Graph it.

Step 1 "At most" means that number or less.

Step 2 Choose open or closed. Then choose the arrow direction.



Inequality_____

5. Find the Error. Keon graphed $n > 5$ with a closed dot at 5 and an arrow to the right. Explain his mistake. Is 5 a solution?

Frame *Keon should use an ___ circle because 5 ___ a solution of $n > 5$.*

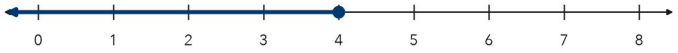
_____

_____

TEST PRACTICE

choose the best answer

6. Which inequality is shown on the number line?



A $x < 4$ **B** $x \leq 4$ **C** $x > 4$ **D** $x \geq 4$

7. Which number is a solution of $n > -3$?

A -5 **B** -4 **C** -3 **D** 0

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___